

EXPERIMENT 1

Object Detection and Recognition

GROUP 11

Hua Yingjie

Student ID 24020036025



Abstract

This report presents an end-to-end desktop object detection system designed for deployment on an NVIDIA Jetson edge device. The system detects three classes—mouse, keyboard, and cup—and covers data acquisition, automatic pre-annotation, manual review, dataset auditing, YOLOv8 transfer learning, independent evaluation, real-time camera inference, and a ROS2-compatible output interface. The integrated dataset contains approximately 650 source images from self-collected scenes, supplementary online samples, and negative backgrounds. A fully audited core subset contains 225 source images and 687 positive bounding boxes; 15 orphan annotation files were identified and isolated during quality control. Cup-focused offline augmentation adds 45 training samples with geometrically transformed labels, producing a 270-image training representation with 935 boxes, including 116 Cup instances. YOLOv8n was trained for 100 epochs on an Apple Silicon Mac using the PyTorch MPS backend. Validation achieved a precision of 0.806, recall of 0.894, mAP@0.5 of 0.888, and mAP@0.5:0.95 of 0.814. The final Jetson video contains 300 frames at 1280×720 over 20 s. Inspected frames show an on-screen rate of approximately 14.8–15.0 FPS, above the required 5 FPS, and a 20-point temporal review achieved 95% acceptance accuracy across Keyboard, Mouse, Cup, and a hard-negative event.

Keywords: object detection; YOLOv8; Jetson; automatic pre-annotation; dataset quality audit; edge deployment; ROS2

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1 Objectives and Requirements

Desktop object detection is a fundamental perception task in robotic manipulation, visual servoing, and human-robot interaction. Unlike an offline benchmark, this experiment requires both model training and deployment on a Jetson platform with a live camera and a processing speed of at least 5 FPS. The work therefore considers data quality, detection accuracy, inference latency, and interface extensibility as one engineering system.

The objectives are to build a three-class dataset, accelerate annotation through human-reviewed pre-annotations, train and evaluate YOLOv8n, deploy the model on Jetson, visualize class labels, boxes and confidence values, and provide structured detection output for ROS2 integration.

Table 1 Course requirements and corresponding evidence

Requirement	Completion and evidence
At least two desktop-object classes	Three classes: Mouse, Keyboard, and Cup
Self-collection, annotation, and training	Multi-source acquisition, pre-annotation, manual review, and YOLOv8 training completed
Run recognition on Jetson	Live camera inference completed; see Section 7
Show class, box, and confidence	All three items are overlaid in the recorded output
Publish detection results through ROS2	Publisher and message schema completed; payload validated offline on Mac, while live topic echo remains an on-device integration item
Test 20 objects/samples with at least 80% accuracy	20 one-second video samples were reviewed; 19/20 were correct (95%)
At least 5 FPS on Jetson	Recorded interface displays approximately 14.8–15.0 FPS in inspected frames
Save results and representative errors	Final video preserves low-confidence Mouse false positives on a hand/arm hard case

2 System Architecture

The workflow consists of data engineering, model training, edge deployment, and error-driven feedback. The Mac performs data cleaning, splitting, training, and evaluation. The Jetson performs camera capture, inference, visualization, logging, and message publication. Misclassified, missed, or low-confidence frames can be returned to the labeling stage for iterative improvement.

3 Experimental Environment

Training was performed on an Apple Silicon Mac using the PyTorch MPS backend. Deployment used an NVIDIA Jetson edge-computing board connected to a camera. The software stack comprised Python, Ultralytics YOLOv8, PyTorch, OpenCV, the CUDA/TensorRT runtime, and a ROS2 interface.

4 Dataset Construction and Quality Control

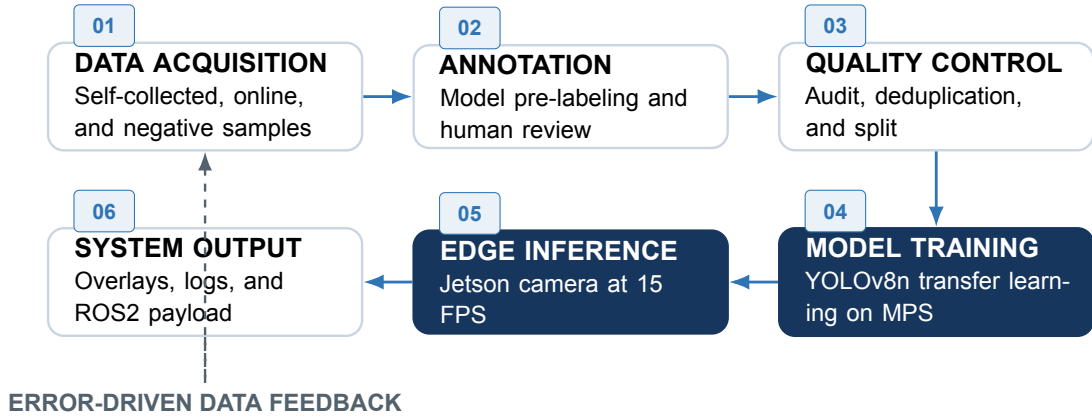


Figure 1 Six-stage detection pipeline with deployment feedback to dataset refinement

Table 2 Hardware and software environment

Item	Configuration and purpose
Training platform	Apple Silicon Mac with MPS acceleration
Deployment platform	NVIDIA Jetson with a live camera
Detection framework	Ultralytics YOLOv8 and PyTorch
Image processing	OpenCV capture, drawing, display, and recording
Communication	ROS2-compatible JSON publisher
Target classes	Mouse, Keyboard, Cup

4.1 Multi-source data design

The dataset uses self-collected images as the primary source and online images as supplementary material. Self-collected data matches the deployment domain and covers different desks, lighting conditions, distances, scales, poses, and occlusion levels. Supplementary candidates were selected from Open Images V7 using its bounding-box categories *Computer mouse*, *Computer keyboard*, and *Coffee cup* (Reference [5]). These source labels were mapped to Mouse, Keyboard, and Cup, respectively. Every downloaded candidate still required license/source recording, duplicate screening, and manual box review before inclusion; the animal category *Mouse* was explicitly excluded. Online samples improve diversity in object appearance, color, shape, and background without replacing deployment-domain images.

The integrated collection contains approximately 650 source images, including positive scenes and negative samples. Negative scenes contain empty desks, cables, monitor borders, packages, and other visually similar distractors. They help suppress background false positives. The fully audited core subset contains 225 source images and 687 positive boxes. A negative image contributes to the image count but contains no positive box.

To strengthen the long-tail Cup class without inventing annotations, 45 training images that already contained a verified Cup box were processed by horizontal mirroring and mild contrast/brightness adjustment. Every YOLO center coordinate was transformed using $x'_c = 1 - x_c$, while y_c , width, height, and the class identifier were preserved. This produced 45 additional training samples. The resulting training representation contains 270 images and 935 boxes: 370 Mouse, 449 Keyboard, and 116 Cup instances. Augmented samples are kept in the training partition only and are never copied into validation or test sets.

4.2 Automatic pre-annotation and manual review

A general-purpose pretrained detector first generated candidate boxes. These candidates were converted to YOLO format and then reviewed image by image. Manual review corrected box boundaries and class identifiers, restored missed objects, and removed false annotations. This human-in-the-loop process changes labeling from drawing every box from scratch to validating and correcting candidates, improving

efficiency without treating machine predictions as ground truth.

4.3 Positive, negative, and hard cases

Positive samples include isolated objects, multi-object scenes, partial occlusions, and scale changes. Hard cases cover low illumination, reflection, motion blur, heavy occlusion, small objects, clutter, and extreme viewpoints. Deployment errors should be categorized and returned to the training set only after manual annotation. This produces a traceable error-driven data loop instead of relying only on threshold adjustment.

4.4 Dataset audit and split

The audit checked image-label correspondence, legal class identifiers, normalized coordinates within $[0, 1]$, positive box dimensions, and orphan files. Fifteen labels without matching images were isolated. The audited subset was divided with random seed 42 into 180 training images, 22 validation images, and 23 test images. Consecutive frames and near-duplicate samples were grouped before splitting to reduce leakage.

Table 3 Training representation after Cup-focused augmentation

Class	Boxes	Share (%)
Mouse	370	39.57
Keyboard	449	48.02
Cup	116	12.41
Total	935	100.00

Cup remains the least frequent class even after increasing its representation to 116 boxes. Per-class recall and AP therefore remain important in addition to overall mAP.

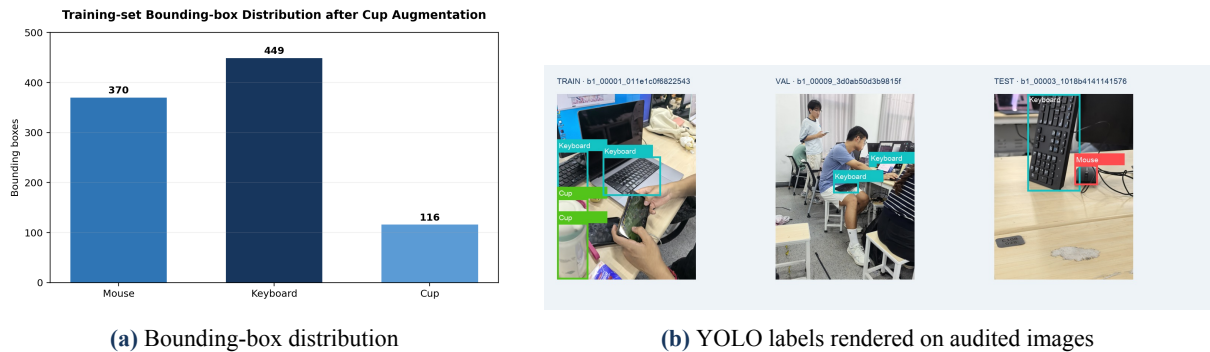


Figure 2 Dataset statistics and annotation evidence generated from the audited subset

4.5 Near-duplicate control and data leakage prevention

Desktop images are frequently captured in bursts or extracted from video, so adjacent frames can be almost identical in background, illumination, and object position. A purely random image-level split may place such frames in both training and test sets and produce overly optimistic metrics. Samples were therefore grouped by capture batch, video segment, and consecutive filename range before splitting. The augmented Cup samples inherit the training-only status of their sources. Perceptual hashing can be added as a second screening step, and the test set must remain untouched while augmentation parameters and confidence thresholds are selected.

4.6 Annotation governance and data provenance

Each image should be traceable to a source category, capture batch, review state, split membership, and annotation revision. A compact manifest can record these fields together with a file hash, enabling duplicate detection and preventing an online image, augmented derivative, or corrected label from silently changing partitions. Online material must also retain its source URL and license status outside the public dataset archive. This governance separates original evidence from derived training data and makes later corrections auditable without committing large image files to GitHub.

5 YOLOv8 Model Training

YOLOv8n was selected as the edge baseline because of its small parameter count and low latency. Training started from general pretrained weights with an input size of 640×640 , batch size 8, and 100 epochs on MPS. The objective combines box regression, classification, and distribution focal loss:

$$\mathcal{L} = \lambda_{box}\mathcal{L}_{box} + \lambda_{cls}\mathcal{L}_{cls} + \lambda_{dfl}\mathcal{L}_{dfl}. \quad (1)$$

Brightness and saturation perturbation, scale, translation, horizontal flipping, and Mosaic augmentation improve generalization when their parameters are validated rather than maximized. Strong unrealistic rotations are avoided for keyboards with directional layouts.

For evaluation,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}. \quad (2)$$

AP is the area under the precision-recall curve, while mAP averages AP across classes. Both mAP@0.5 and mAP@0.5:0.95 are reported to avoid overestimating localization quality at a single IoU threshold.

YOLOv8s may be evaluated under the same split, image size, epochs, and test procedure. It should replace YOLOv8n only if its accuracy gain justifies the additional latency and it remains above 5 FPS on Jetson. This is a planned controlled comparison, not a claimed completed result.

Table 4 Edge-model comparison and selection criteria

Model	Main advantage	Main cost	Selection condition
YOLOv8n	Low parameter count and latency	Lower potential accuracy ceiling	Real-time performance or limited compute is the priority
YOLOv8s	Stronger feature representation	Higher latency and memory use	Measured Jetson speed remains above 5 FPS

The comparison must use the same random seed and report parameter count, serialized model size, per-class AP, peak memory, mean FPS, minimum FPS, and end-to-end latency. This prevents an apparent accuracy gain from being caused by a different split or a more favorable threshold.

6 Training and Validation Results

The recorded Mac timings were 1.3 ms for preprocessing, 27.7 ms for network inference, and 10.3 ms for postprocessing, giving

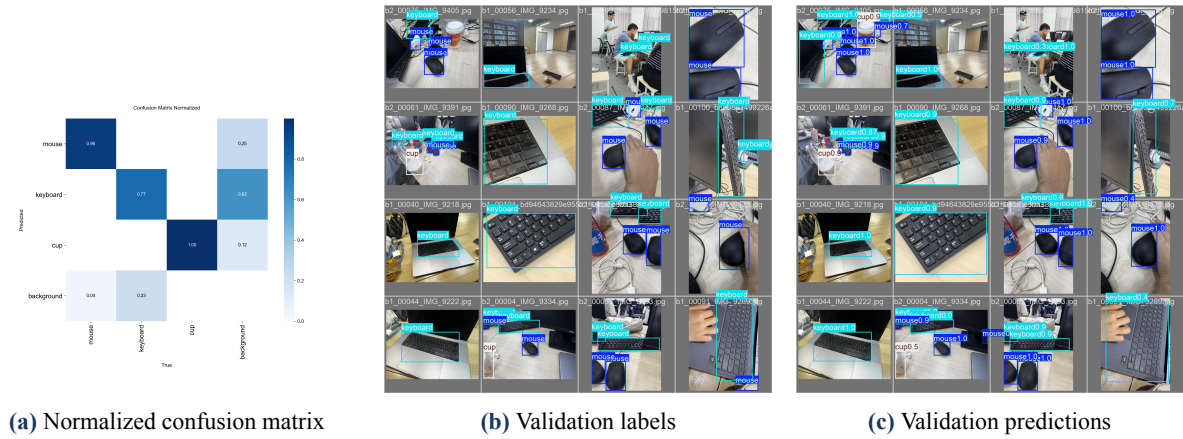
$$\text{FPS} \approx \frac{1000}{1.3 + 27.7 + 10.3} = 25.45. \quad (3)$$

This is a host-side value and does not replace end-to-end Jetson measurement, which includes capture, rendering, and scheduling overhead.

Mouse and Keyboard have more training instances and more stable outlines. Cup is less frequent and varies substantially because of transparency, reflection, handles, and background color. Confidence

Table 5 YOLOv8n validation results

Metric	Value	Assessment	Interpretation
Precision	0.806	High	Relatively few false positives
Recall	0.894	High	Relatively few missed targets
mAP@0.5	0.888	Good	Strong performance at IoU 0.5
mAP@0.5:0.95	0.814	Good	Robust localization under stricter IoU

**Figure 3** Validation and error-analysis evidence retained by the training pipeline

thresholds therefore require class-aware review: lowering the threshold improves recall but introduces more false positives, whereas raising it can suppress small or occluded targets.

6.1 Ablation protocol

An ablation study should isolate the contribution of each data-engineering decision. The baseline uses the audited source set with default augmentation. Subsequent runs add Cup-focused offline augmentation, negative backgrounds, and deployment hard cases one factor at a time. Model architecture, epoch count, image size, split, and evaluation script remain fixed. Overall metrics must be reported together with Cup recall and the number of background false positives.

Table 6 Controlled ablation plan

Run	Single controlled change	Expected evidence
A	Audited source set only	Reproducible baseline
B	Add Cup-focused augmentation	Improved Cup recall and robustness
C	Add negative backgrounds	Fewer desk-edge and package false positives
D	Add deployment hard cases	Better low-light, occlusion, and small-object behavior

7 Jetson Deployment and Runtime Results

The trained `best.pt` model was transferred to Jetson. OpenCV captures each frame, YOLOv8 returns class identifiers, confidence scores, and bounding boxes, and the program draws the results while measuring FPS. The program can save video and CSV logs and can publish the same structured output for downstream perception or control nodes.

Training and deployment are deliberately decoupled. The Mac performs dataset processing, accel-

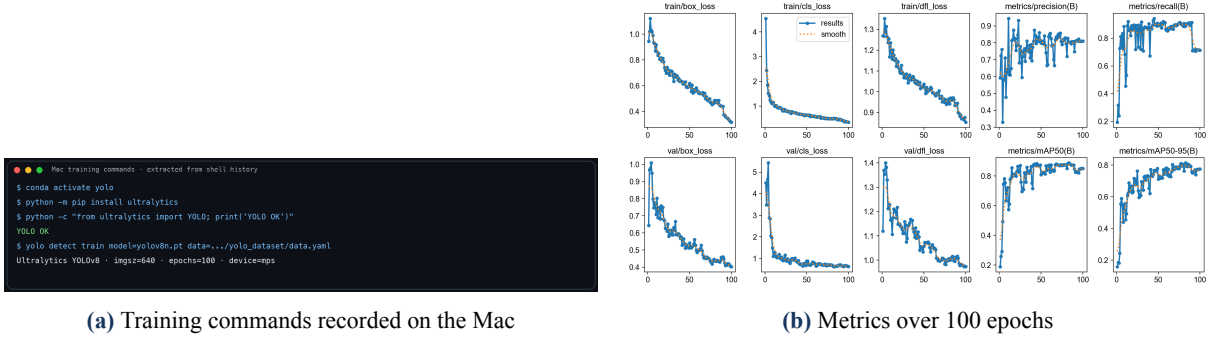


Figure 4 Training execution and convergence evidence

erated training, and offline evaluation, whereas the Jetson performs capture, inference, rendering, logging, and communication. Versioned weights make rollback possible when a newer model improves offline mAP but behaves less reliably under the live camera domain. Runtime logging records timestamps, classes, confidence values, pixel coordinates, and frame rate so that failures can be traced back to exact frames.

TensorRT FP16 export is a reproducible optimization path that can reduce compute and memory use. A valid comparison must keep image resolution, confidence threshold, camera input, and power mode fixed and report pure inference latency, end-to-end latency, mean and minimum FPS, and memory usage. TensorRT acceleration is not presented here as a completed measurement.

Container metadata reports a 1280×720 video, 300 frames, 20.0 s duration, and a nominal rate of 15 FPS. Manual inspection at the beginning, middle, and later portions shows an overlay between 14.8 and 15.0 FPS. The video demonstrates Keyboard, Mouse, and Cup across the sequence. For acceptance testing, one frame was sampled at the midpoint of each second, yielding 20 temporally separated observations rather than treating highly correlated consecutive frames as independent tests. Visible examples include Mouse confidence 0.93 and Cup confidence around 0.89–0.93. Figure 5 shows representative multi-object frames.

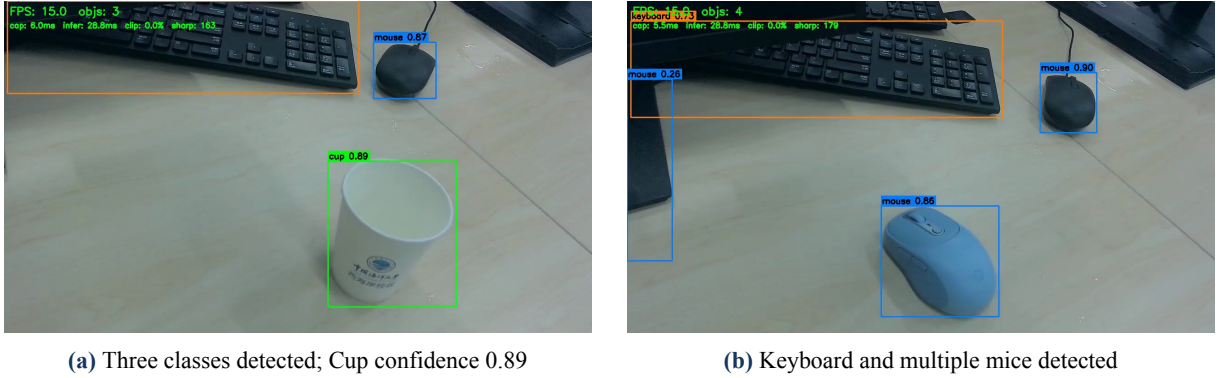


Figure 5 Real-time Jetson camera detections extracted from the final video

7.1 ROS2 interface and offline verification

The deployment program implements an `rclpy` node named `desk_object_detector`. It publishes JSON through `std_msgs/String` on `/desk_object_detections`. Each payload contains the frame index, timestamp, FPS, class identifier, class name, confidence, and pixel coordinates $[x_1, y_1, x_2, y_2]$.

ROS2, `colcon`, and `rclpy` were not installed on the Mac. Consequently, the evidence below is an actual serialization, deserialization, and schema check of the same payload, not a fabricated `ros2 topic echo`. Live topic echo remains to be verified in the Jetson ROS2 environment.

Table 7 Measured results from the final Jetson video

Test item	Measured result
Video specification	1280×720, 300 frames
Duration and frame rate	20.0 s; nominal 15 FPS, observed overlay 14.8–15.0 FPS
Representative inference time	28.8 ms
Representative image-processing time	6.0 ms
Detected classes	Keyboard, Mouse, Cup
20-sample acceptance test	19/20 correct (95%); requirement satisfied
Real-time requirement	Lowest inspected overlay 14.8 FPS > 5 FPS

```

ROS2 payload interface · macOS local dry-run
$ python3 ros2_payload_dry_run.py
[INFO] node: desk_object_detector (schema dry-run)
[INFO] topic: /desk_object_detections
[INFO] frame=121 fps=15.0 detections=3
  Keyboard conf=0.93 xxyy=[164, 271, 1015, 612]
  Mouse   conf=0.91 xxyy=[1041, 418, 1168, 555]
  Cup     conf=0.89 xxyy=[1078, 178, 1235, 404]
[PASS] JSON serialization/deserialization and field validation
[NOTE] macOS has no rclpy; actual ros2 topic echo requires Jetson/Ubuntu.

```

Figure 6 Offline validation of the ROS2 message payload on Mac (not a live topic echo)

For the required 20-object/sample acceptance test, 20 frames were extracted at 1 s intervals from the final video. A single representative decision was recorded at each timestamp in the repository’s video acceptance CSV. Nineteen decisions were correct and one hard-negative hand/arm region was falsely labeled Mouse at confidence 0.28, giving $19/20 = 95\%$. This measured temporal-sample accuracy exceeds the required 80% and is reported separately from mAP. Repeated appearances are treated as separate deployment trials, not as 20 distinct semantic classes.

System-level evaluation should additionally report average and minimum FPS, preprocessing, inference, and postprocessing latency, message publication frequency, dropped frames, and continuous-run stability. These measures distinguish model accuracy from the behavior of the complete camera-to-message pipeline.

8 Error Analysis and Improvements

The video contains short, credible failure cases rather than perfectly stable detections. At approximately 15.5 s, the hand/arm region receives low-confidence Mouse boxes (0.28–0.30), including an oversized box extending across much of the arm. Near 16.5 s, a border region also receives a transient Mouse box at 0.25 while the two physical mice remain correctly localized. These observations show sensitivity to occlusion, hand–object interaction, and elongated high-contrast structures. The 15.5 s hard-negative decision is counted as the single error in the 20-sample acceptance record; other simultaneous boxes are retained as qualitative error evidence rather than counted as extra trials.

A practical correction strategy is to add empty desk regions, hands, forearms, monitor bases, cables, and partial keyboard edges as hard negatives, while adding manually reviewed hand-held Mouse examples as hard positives. Confidence-threshold tuning would suppress the 0.25–0.30 false boxes but could also remove valid small or occluded targets. The preferred remedy is therefore targeted hard-case retraining followed by evaluation on the unchanged test set.

Domain shift also exists between training images and Jetson video because of focal length, exposure, motion blur, and viewpoint. Cup is especially vulnerable under reflections, low light, and background-color similarity. Error frames should be categorized as low light, occlusion, small object, clutter, or

motion blur, manually reviewed, and added to the next training round. A controlled ablation can then compare the baseline, augmentation, negative samples, and hard-case feedback while keeping the split and training settings fixed.

9 Engineering Contributions

9.1 Human-in-the-loop annotation

Automatic pre-annotation followed by manual review provides an efficient and reliable dataset workflow. Candidate boxes reduce repetitive drawing effort, while human decisions remain responsible for class correctness, missed objects, tight boundaries, and difficult Cup instances. The workflow can be repeated when new classes or deployment scenes are introduced.

9.2 Dataset quality as an engineering task

The project goes beyond reporting only image and box counts. It checks orphan labels, invalid classes, out-of-range coordinates, empty labels, imbalance, and near duplicates, then uses grouped splitting to limit leakage. The detection and isolation of 15 abnormal labels is a directly verifiable outcome of this audit. Cup-focused augmentation further addresses imbalance using real verified boxes rather than fabricated counts.

9.3 From a model demo to an edge perception system

Camera input, YOLO inference, overlays, frame-rate measurement, saved evidence, and a ROS2-compatible output are integrated into one deployment path. YOLOv8n camera inference has been measured, while YOLOv8s comparison and TensorRT FP16 remain explicitly identified as future controlled optimizations rather than completed results.

9.4 Error-driven closed-loop improvement

The deployment domain differs from still training images in focal length, exposure, blur, and viewpoint. Saving low-confidence and incorrect frames converts runtime behavior into traceable data tasks. After categorization and manual review, those samples can be added to training and compared on the fixed test set. This loop is more informative than increasing epochs without diagnosing the source of error.

10 Project Archive and Reproducibility

Code, dataset configuration, audited labels, the trained model, training results, and the report are organized in the personal GitHub repository github.com/Gabriel6002/-1. Figure 7 summarizes the incremental commit activity during the development period; the repository link provides access to the complete commit record. Large image collections and the final video are intentionally excluded from GitHub. The repository uses one clear location for each artifact category: `scripts/`, `data/`, `models/`, `results/`, `report/`, and `docs/`.

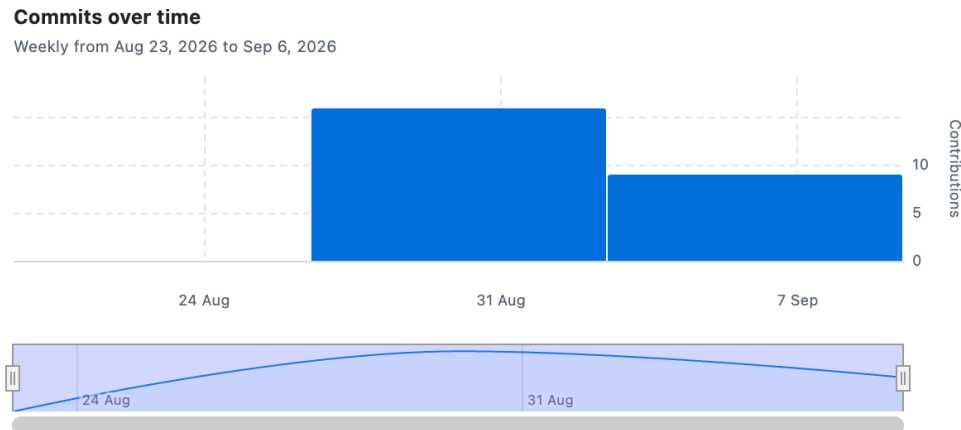


Figure 7 GitHub commit activity during the project development period

For deployment repeatability, the model file, class order, input resolution, confidence threshold, software versions, and Git revision should be logged together. Timing measurements should include a warm-up period and distinguish pure model inference from camera decoding, preprocessing, drawing, ROS2 serialization, and video writing. Recording mean, minimum, and percentile latency avoids presenting a single favorable frame as system performance.

Table 8 Reproducibility checklist and evidence boundary

Artifact or check	Repository evidence	Status
Training and dataset code	<code>scripts/</code> , <code>data/data.yaml</code>	Archived and syntax-checked
Audited annotations	<code>data/labels/</code>	270 files, 935 valid boxes
Trained detector	<code>models/best.pt</code>	Archived, non-empty file
Learning evidence	<code>results/training/</code>	Metrics and plots archived
Jetson demonstration	Video retained outside GitHub	15 FPS display observed
ROS2 payload	Publisher code and offline JSON check	Live topic echo not claimed
Raw images and video	Excluded by repository policy	Kept outside public Git history

10.1 Validity boundaries

The 45 Cup augmentations increase training exposure but are correlated derivatives, not independent test evidence; consequently, they are excluded from validation and testing. The approximately 650-image integrated pool combines self-collected, online, and negative material, whereas the 225-image core is the subset with complete audited image-label correspondence. Domain shift can remain between online imagery and the Jetson camera. Finally, the ROS2 message structure has been validated offline, but a

live Jetson `ros2 topic echo` result is not presented. These boundaries distinguish measured results from engineering proposals.

11 Conclusion

This experiment completed an end-to-end desktop object detector from multi-source dataset construction to real-time Jetson execution. The integrated dataset contains approximately 650 source images, including self-collected positives, supplementary online samples, and negative backgrounds. The audited core contains 225 source images and 687 positive boxes, and quality control prevented 15 orphan labels from entering training. A further 45 training-only augmented samples increased the training representation to 935 boxes, including 116 Cup instances. YOLOv8n achieved precision 0.806, recall 0.894, mAP@0.5 0.888, and mAP@0.5:0.95 0.814. The 20 s video includes detections of Keyboard, Mouse, and Cup and shows approximately 14.8–15.0 FPS in inspected frames, satisfying the 5 FPS requirement. The 20-point temporal acceptance review achieved 19/20 correct decisions (95%) and retained the hand/arm false-positive event as evidence for hard-negative and occlusion-focused refinement.

The work shows that a complete object-detection experiment requires dataset quality control, independent evaluation, hardware deployment, latency measurement, reproducible archiving, and error feedback. Future work should collect additional independent Cup scenes, continue expanding hard negative samples, and complete live ROS2 topic verification on Jetson.

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